

Antifouling Property of Cu₂O-Free Self-Polishing Antifouling Coatings Based on Amide Derivatives Inspired by Capsaicin

Xuan Wang, Jian Yang, Zhenxia Liu, Xiaohui Jiang,* and Liangmin Yu*



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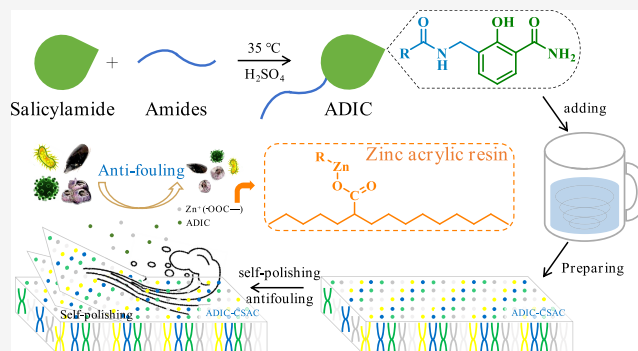


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ABSTRACT: The evidence from many studies shows that antifoulants (Cu₂O) and organic antifouling agents with broad-spectrum characteristics in antifouling coatings cause varying degrees of damage to the environment. Therefore, this study prepared Cu₂O-free self-polishing antifouling coatings based on amide derivatives inspired by capsaicin (ADIC-CSAC) with green and environmentally friendly characteristics. First, the structure of ADIC and the composition of ADIC-CSAC were characterized by IR, ¹H NMR, ¹³C NMR, HRMS, and EDX. Moreover, antibacterial, anti-algal, static raft tests and changes in the mass loss, roughness, contact angle, and surface energy were used to evaluate the antifouling and self-polishing properties of ADIC-CSAC. The test results showed that ADIC and ADIC-CSAC were successfully prepared and ADIC-CSAC possessed good antifouling and self-polishing properties. ADIC-CSAC exhibited antibacterial and anti-algal rates of over 88 and 72%, respectively, and was found to have satisfactory antifouling properties over 9 months in a real marine field. Overall, the prepared ADIC-CSAC possesses good and green antifouling and self-polishing properties, which lays a foundation for research on green antifouling coatings used for environmental protection.



INTRODUCTION

The ocean provides humans with resources and energy and plays an important role in economic interests and strategic national defense.^{1–3} Marine organisms are inevitably negatively affected by the development of the marine economy.^{4–6} Biological dirt formed by the adsorption, growth, and reproduction of marine microorganisms, animals, and plants on the surface of marine facilities results in enormous economic losses and is a potential safety hazard.^{7–9} Many methods have been developed to prevent marine pollution, such as mechanical removal, direct addition of toxins to seawater, and besmearing surfaces with antifouling coatings.^{10–12} Besmearing surfaces with antifouling coatings is currently recognized as the most economical, effective, and widely used numerous antifouling technologies.^{13,14} Mainstream antifouling coatings that are currently available worldwide are self-polishing antifouling coatings with zinc acrylate resin as the film-forming material, Cu₂O as the main antifouling agent, and an organic bactericide as the auxiliary antifouling agent.^{15–17} However, a growing demand worldwide for environmental and ecological protection has focused increasing attention on the environmental harm caused by Cu₂O. The toxicity of the key ingredient Cu₂O and the fast release of bactericide for traditional copper-based antifouling coatings are two major sources of severe environmental risk that restrict future widespread use.^{18,19} It has also been confirmed in studies that copper ions accumulate in marine

organisms.^{20–22} Therefore, many countries have included Cu₂O antifouling coatings in a list of substances associated with “high pollution and high environmental risk”, and the use of Cu₂O in yacht antifouling coatings has been explicitly prohibited on the east coast of Sweden.²³ Therefore, there is a general trend toward developing environmentally friendly antifouling coatings that do not contain copper or heavy metal fungicides.

The greatest strength of copper-free antifouling coatings is enhanced environmental protection, and these coatings are already being produced by some large foreign companies. However, these coatings also have significant disadvantages.²⁴ First, these coatings are very expensive. Second, the antifouling effect and broad spectrum are not comparable to those in which Cu₂O is the main antifouling agent. A variety of organic antifouling agents often need to be combined with copper-free coatings to ensure an antifouling effect. Therefore, current research on copper-free antifouling coatings is mainly focused on finding efficient, nontoxic, and environmentally friendly

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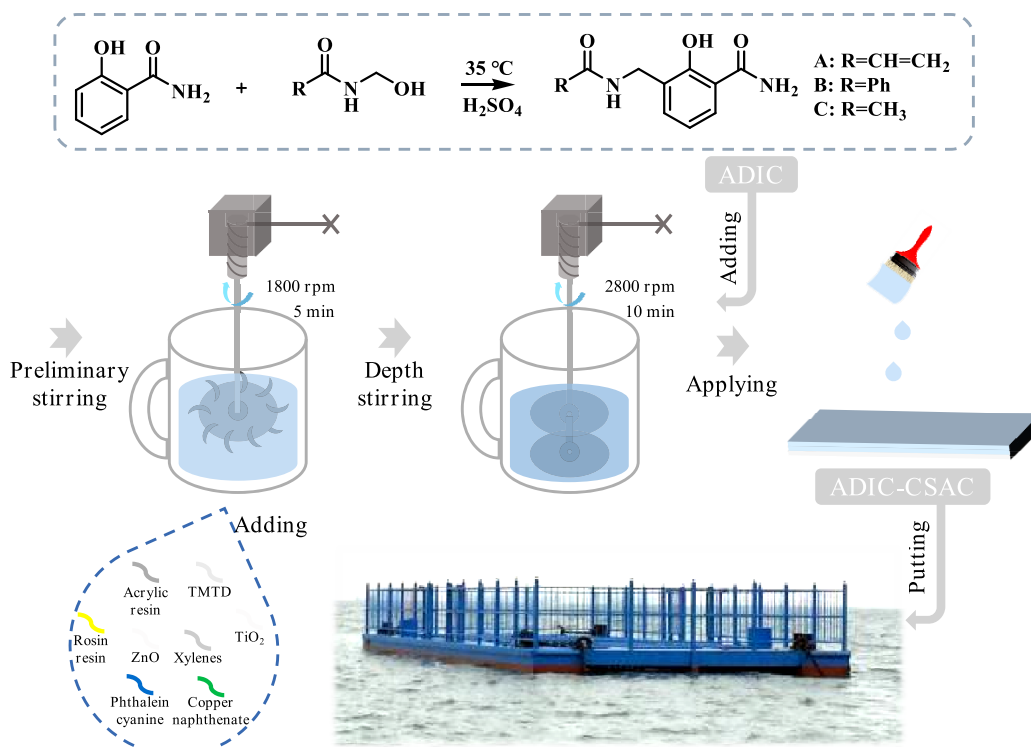


Figure 1. Preparation of ADIC-CSAC.

organic antifouling agents that can prevent biological adhesion without damaging the environment.^{22,25–27} For example, polyphenol compounds extracted from brown algae, camptothecin extracted from *Camptotheca acuminata*, gingerol obtained from ginger, and capsaicin extracted from spicy plants can prevent the attachment and growth of marine organisms to a surface, resulting in high antifouling activity.²⁸ However, the complex process required to extract a natural antifouling agent from organisms, poor compatibility with the matrix resin, and the sustenance of stable and controllable release are unresolved problems, such that commercial application remains distant. Therefore, a variety of synthetic green antifouling agents have been developed. One novel agent is 2-(*p*-chlorophenyl)-3-cyano-4-bromo-5-trifluoromethyl-pyrrole, which is synthesized based on the structural bionics of metabolites of microbial *Streptomyces* that has broad spectrum and excellent antifouling activity for marine fouling invertebrates.^{29,30} Capsaicin derivatives with excellent antifouling effects can also be obtained manually in high yield and purity. The continuous development of synthetic green antifouling agents for environmental protection is a step toward commercial application and furthers the development process for replacing Cu₂O as the main antifouling agent.

The purpose of this study was to obtain a coating formulation that could match the antifouling effect of traditional antifouling coatings containing Cu₂O while being friendly to the marine environment. Therefore, amide derivatives inspired by capsaicin (ADIC) as antifouling agents were compounded with zinc acrylate resin as a self-polishing monomer to prepare environmentally friendly Cu₂O-free dual-functional coatings (Cu₂O-free self-polishing antifouling coatings based on ADIC (ADIC-CSAC), Figure 1). Antibacterial and anti-algal assays were used to evaluate the antibacterial and anti-algal properties of ADIC-CSAC against *Escherichia coli* (*E. coli*), *Staphylococcus aureus* (*S. aureus*), *Phaeodactylum*

tricornutum (*P. tricornutum*), *Chaetoceros curvisetus* (*C. curvisetus*), *Skeletonema costatum* (*S. costatum*), and *Nitzschia closterium* (*N. closterium*). The antifouling activities of ADIC-CSAC were determined by a marine field test. The self-polishing, smoothness, and wettability of the coating were characterized using static state simulation assays, roughness (Ra) measurements, and contact angle (CA) measurements, respectively. The ADIC structure and chemical compositions of ADIC-CSAC were determined by infrared spectroscopy (IR), nuclear magnetic resonance spectroscopy (¹H NMR and ¹³C NMR), high-resolution mass spectrometry (HRMS), and energy-dispersive X-ray spectroscopy (EDX). The results indicated the feasibility of preparing ADIC-CSAC as a dual-functional coating to inhibit biofilm formation, and the lack of Cu₂O in the coating is expected to significantly reduce pollution of the aquatic environment by the heavy metal copper, fuel consumption of ships, and emissions of CO₂ and SO₂, which is in line with national energy conservation and emission reduction policies and extremely important for environmental and ecological protection.

EXPERIMENTAL SECTION

Materials. Salicylamide, ethyl alcohol, and concentrated sulfuric acid were obtained from Sinopharm Chemical Reagent Co., Ltd. (Beijing, China), and *N*-hydroxymethylacrylamide (N1), *N*-hydroxymethylbenzamide (N2), and *N*-hydroxymethylacetamide (N3) were obtained from Zibo Zhongsen Chemical Co., Ltd. (Shandong, China), Beijing Chengyu Specialty Chemical Co., Ltd. (Beijing, China), and Shanghai Yaxing Biomedical Technology Co., Ltd. (Shanghai, China), respectively. *E. coli*, *S. aureus*, *P. tricornutum*, *C. curvisetus*, *S. costatum*, and *N. closterium* were provided by the Ocean University of China (Qingdao, China).

Preparation of Coatings. ADIC-CSAC was prepared according to a method reported by Wang et al.³¹ Efficient green organic antifouling coatings were configured using zinc acrylate resin and ADIC as the antifouling agent. ADIC (7.0 g), zinc acrylic resin (21.1

g), a TMTD accelerator (15.6 g), ZnO (15.6 g), rosin resin (20.2 g), copper naphthenate (4.7 g), TiO₂ (11.6 g), phthalein cyanine (0.3 g), and xylene (3.9 g) were mixed; the mixture was then stirred at 1800 rpm for 5 min with a stainless steel stirrer and at 2800 rpm for 10 min with a plastic stirrer to yield uniform coatings. A coating based on chlorothalonil (CT-CSAC) was used as a control. ADIC was synthesized by a Friedel–Crafts (F-C) alkylation reaction. First, 0.1 mol of salicylamide and 0.12 mol of different amides (N1, N2, and N3) were added to a three-necked flask and completely mixed with absolute alcohol (100 mL). Concentrated sulfuric acid (10 mL) was carefully added to the flask, and the mixture was stirred at 35–40 °C for up to 72 h. ADIC (*N*-(2-hydroxy-3-amidebenzyl) acrylamide (A), *N*-(2-hydroxy-3-amidebenzyl) benzamide (B), and *N*-(2-hydroxy-3-amidebenzyl) acetamide (C)) were obtained after multiple recrystallizations; the ADIC characterization results are shown in Table 1, Figure 2, and Figures S1 and S2.

Characterization. An IR (TENSOR II Fourier transform infrared spectrometer, Bruker, Germany) was used to confirm the structure and chemical compositions of ADIC and ADIC-CSAC. ¹H NMR (Bruker Avance spectrometer, Bruker, Switzerland), ¹³C NMR (Bruker Avance spectrometer, Bruker, Switzerland), and HRMS (Agilent 6530 Accurate-Mass Q-TOF LC/MS, Agilent, USA) were employed to further confirm the ADIC structure. EDX (Apollo XLT SDD, EDAX, USA) was employed to determine the surface chemical compositions of ADIC-CSAC.

Thermogravimetric (TG) Assay. Three milligrams of dried ADIC or ADIC-CSAC was placed in a crucible built into a thermal gravimetric analyzer (STA449C, NETZSCH, Germany) to measure the TG and derivative thermogravimetric (DTG) properties of the materials.^{32,33} The following conditions were used: the gas environment was nitrogen, the gas flow was 40 mL·min^{−1}, the temperature range was 30–800 °C, the heating rate was 10 °C·min^{−1}, and the holding time was 1 min.

Static State Simulation Assay. A static state simulation assay was used to carry out hydrolytic degradation tests on the coatings.^{34,35} First, numbered and dried PVC panels and PVC panels covered with the coatings were weighed to obtain the masses *m* and *M*₀. The coated PVC panels were then soaked in natural seawater for 50 days, and a cycle was performed every 10 days. After each cycle, the coated PVC panels were removed from the seawater, cleaned, dried, and weighed to obtain the mass *M*_v and the mass loss of the coatings in seawater with time was calculated using formula 1.

$$\text{mass loss} = (M_0 - M_v) / (M_0 - m) \times 100\% \quad (1)$$

In addition, the surface Ra and CAs (the WCA (water contact angle) and DCA (diiodomethane contact angle)) of the coatings were measured by laser scanning confocal microscopy (LSCM, OLS4000, Olympus, Beijing) and a contact angle system (DSA100, KRÜSS, Germany). Three parallel samples were tested for each coating, and the average of measurements at 5 points on each parallel sample was recorded. The surface energy (SE) was calculated by the Owens–Wendt–Rabel–Kaelble (OWRK) geometric average method using water and diiodomethane, which surface energy parameters are shown in Table S1.^{36–39} The calculation formulas are as follows:

$$\gamma_s = \gamma_s^d + \gamma_s^p \quad (2)$$

$$\gamma_l(1 + \cos \theta) = 2(\gamma_s^d \gamma_l^d)^{1/2} + 2(\gamma_s^p \gamma_l^p)^{1/2} \quad (3)$$

where γ_s , γ_s^d , γ_s^p , γ_l , γ_l^d , and γ_l^p are the SE of the coating, the dispersive part of the SE of the coating, the polar part of the SE of the coating, the standard liquid SE, the dispersive part of the SE of the liquid surface, and the polarity part of the SE of the liquid surface, respectively.

Antibacterial Test. The antibacterial activities of the ADIC-CSAC coatings against two types of bacteria were examined using the colony count method.^{40–42} *E. coli* and *S. aureus* are common representative Gram-negative and Gram-positive bacteria, respectively. Accordingly, these bacteria are recommended for antibacterial testing. In preparation for the antibacterial tests, the bacterial strains

Table 1. ADIC Characterization Results

ADIC	A	B	C
feature	white solid	white solid	white solid
m.p./°C	170.2–171.5	147.2–149.3	196.3–198.2
yield	0.84	0.82	0.75
IR (KBr), ν	755.3, 1118.36, 1232.09, 1535.61, 1598.88, 1637.29, 1679.47, 3079.62, 3298.79, 3371.10	689.92, 1033.86, 1108.87, 1540.58, 1598.04, 1629.16, 1646.72, 2969.79, 3057.57, 3295.37	759.85, 1113.84, 1236.60, 1540.13, 1586.08, 1641.06, 1670.43, 3075.86, 3328.92, 3366.58
¹ H NMR (600 MHz,DMSO- <i>d</i> ₆), δ	4.73 (s, 2H, CH ₂), 5.64 (m, 1H, =CH), 6.16 (m, 1H, =CH ₂), 6.29 (m, 1H, =CH ₂), 6.89 (m, 1H, PhH), 6.92 (m, 1H, PhH), 7.41 (m, 1H, PhH), 7.91 (m, 1H, NH), 8.81 (t, 1H, NH), 9.38 (t, 1H, NH), 12.27 (s, 1H, OH)	4.89 (t, 2H, CH ₂), 6.89 (d, 1H, PhH), 6.92 (d, 1H, PhH), 7.40 (m, 1H, PhH), 7.47 (s, 1H, PhH), 7.54 (m, 1H, PhH), 7.90 (s, 1H, PhH), 7.91 (s, 1H, PhH), 7.94 (m, 1H, PhH), 9.05 (t, 1H, NH), 9.18 (t, 1H, NH), 9.33 (t, 1H, NH), 12.22 (s, 1H, OH)	1.84 (s, 3H, CH ₃), 4.62 (t, 2H, CH ₂), 6.89 (m, 1H, PhH), 6.92 (d, 1H, PhH), 7.40 (d, 1H, PhH), 7.90 (t, 1H, NH), 8.57 (t, 1H, NH), 9.31 (t, 1H, NH), 12.30 (s, 1H, OH)
¹³ C NMR (150 MHz,DMSO- <i>d</i> ₆), δ	44.35, 115.85, 117.77, 119.31, 126.58, 128.58, 131.80, 134.39, 159.97, 165.41, 169.27	45.26, 116.22, 117.70, 119.31, 127.88, 127.90, 128.72, 128.79, 129.26, 132.01, 134.18, 134.26, 159.69, 167.21, 168.71	22.89, 44.27, 115.84, 117.76, 119.27, 128.94, 134.35, 160.02, 169.24, 170.35
HRMS/ <i>m/z</i> (calcd)	221.0928	271.1080	209.0893
	[M + H] ⁺ : 221.2274; 243.0753	[M + H] ⁺ : 271.2448; 293.0900	[M + H] ⁺ : 209.2218; 231.0747
	[M + Na] ⁺ : 243.2196	[M + Na] ⁺ : 293.0902	[M + Na] ⁺ : 231.2140

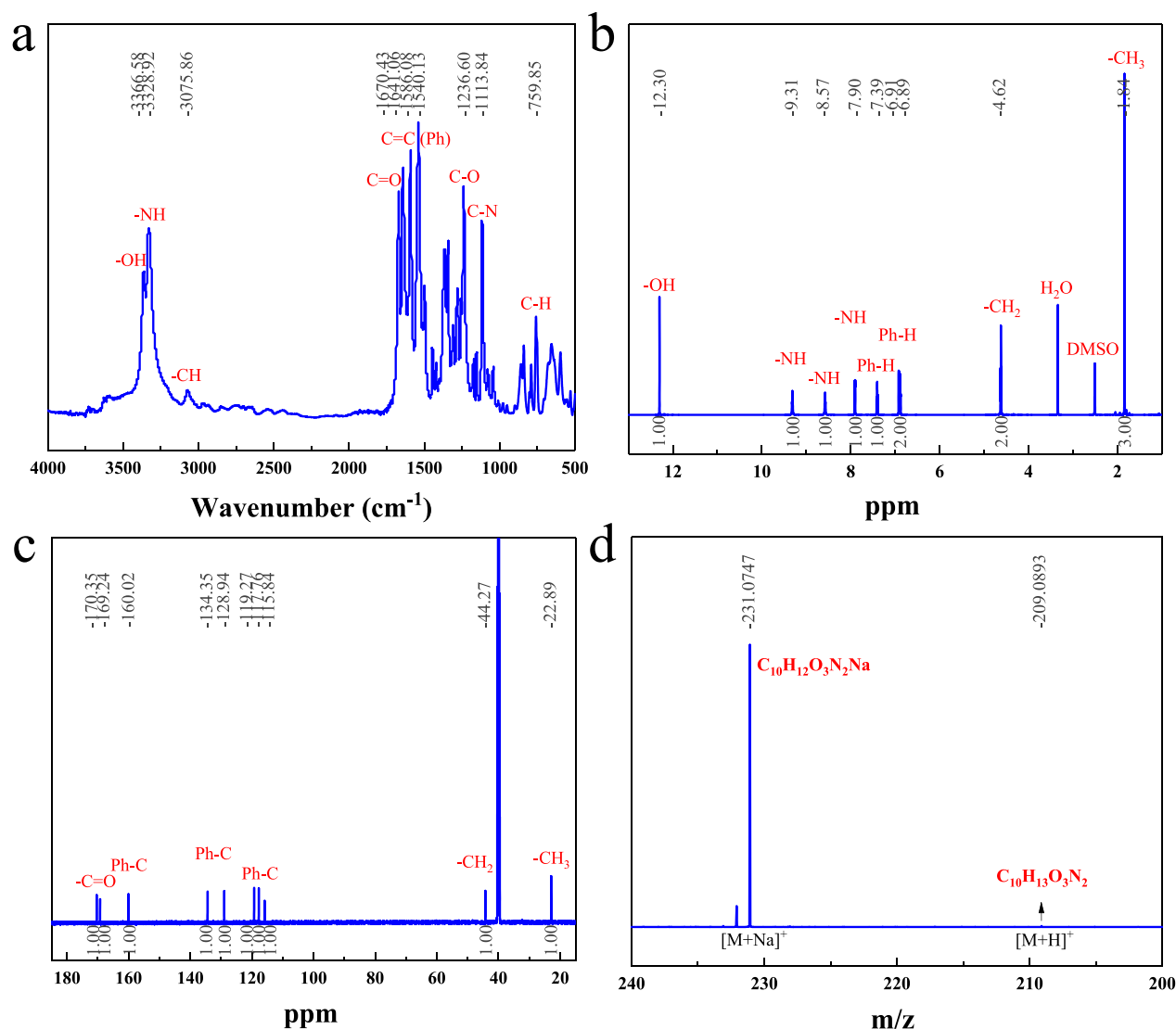


Figure 2. IR (A), ¹H NMR (B), ¹³C NMR (C), and HRMS (D) of Compound C.

were activated at 37 °C and then cultured for 24 h at 37 °C in nutrient broth until the bacterial growth entered the exponential phase. Then, 100 μL of a bacterial suspension was dripped uniformly over the surface of an ADIC-CSAC coating (1 × 1 cm²). The sample was left in an incubator for 18 h, the surviving bacteria on the coating surface were washed with sterile saline (5 mL), and then cultured on nutrient agar for 18 h. The antibacterial rate was calculated using formula 4

$$A = (A_0 - A_i) / A_0 \times 100\% \quad (4)$$

where A represents the antibacterial rate of ADIC-CSAC and A_0 and A_i represent the number of bacterial colonies of the blank and test panels, respectively.

Anti-Algal Test. The anti-algal activities of ADIC-CSAC against a few types of diatoms were measured using a spectrophotometer. *P. tricornutum*, *C. curvisetus*, *S. costatum*, and *N. closterium* are common representative diatoms that can easily cause red tide.^{43,44} In preparation for the anti-algal tests, the algal strains were cultured for several days in natural seawater until the algal growth entered the exponential phase and then diluted to a concentration of 10⁻⁵ mL⁻¹. ADIC-CSAC (5 × 5 cm²) was added to 50 mL of an algal suspension and cultured for 7 days. The absorbance of the algal solution was measured every day at the algal maximum absorption wavelength (Table S1), and the concentration of the algal solution was determined by using standard working equations (Table S2). All

the abovementioned algal solutions were cultured in a climatic cabinet with a 12:12 h L/D cycle at 20 °C.

Marine Field Test. A marine field test on ADIC-CSAC was performed following a previously reported experimental method over 9 months at Qingdao Port (36° 04' N, 120° 19' E) in China, which is extremely rich in marine biological resources.⁴⁵ Polyvinyl chloride (PVC) panels covered with ADIC-CSAC were immersed at depths of 0.2–2 m and spacings of more than 200 mm. After immersion in the sea, the samples were inspected regularly as needed, and the observation time was shortened as much as possible. After observation, each sample was immediately re-immersed in the sea to prevent the death of the attached organisms and a subsequent effect on the experimental results.

RESULTS AND DISCUSSION

Characterization Results of ADIC and ADIC-CSAC. F-C alkylation is a simple, economical, and green method for the efficient embedment of amides in aromatic compounds; this reaction can be used to prepare ADIC with good antibacterial and anti-algal activities, which can then be properly added to coatings to improve the antifouling activity. The characterization results of ADIC and ADIC-CSAC were analyzed in detail, where the spectra for Compounds C and C-CSAC are shown as representative results in Figures 2 and 3.

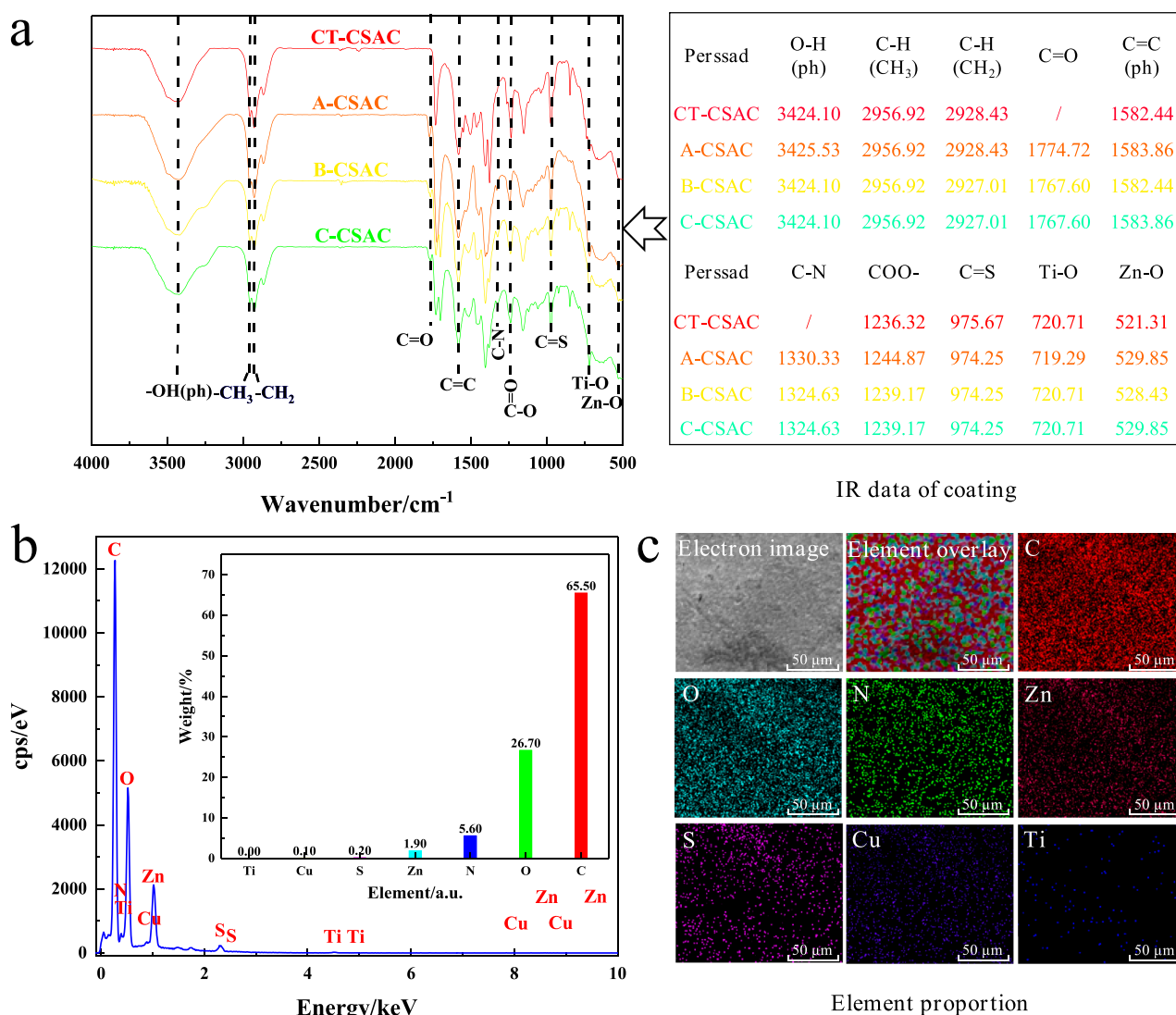


Figure 3. IR spectra of the coatings and surface chemical composition analysis of C-CSAC: IR spectra of the coatings (a), EDX spectrum and elemental mass ratio of C-CSAC (b), and EDX element mapping for C, O, N, Zn, S, Cu, and Ti (c).

Figure 2 shows (a) peaks corresponding to the CH out-of-plane bending and C=C stretching modes of the benzene ring at 759.85 and 1540.13 (1586.08 and 1641.06) cm^{-1} ; peaks for the C-N, C=O, and -NH- (NH_2) stretching modes of the amide at 1113.84, 1670.43, and 3328.92 cm^{-1} , respectively; peaks for the C-O and -OH stretching modes of the phenolic hydroxyl at 1236.60 and 3366.58 cm^{-1} , respectively; and peaks for the CH stretching mode of the methylene at 3075.86 cm^{-1} ; (b) peaks of the Hs of the methyl, methylene, benzene ring, amide, and phenolic hydroxyl at 1.84, 4.62, 6.89 (6.91 and 7.39), 7.90 (8.57 and 9.31), and 12.30 ppm, respectively; (c) the peaks of the C of the methyl, methylene, benzene ring, and amide at 22.89, 44.27, 115.84 (117.76, 119.2, 128.94, 134.35, and 160.02), and 169.24 (170.35) ppm, respectively; and (d) the calculated molecular weight of Compound C is 208.2086, which is in agreement with the experimental values of 209.0893 $[\text{M} + \text{H}]^+$ and 231.0747 $[\text{M} + \text{Na}]^+$.

In addition, Figure 3 shows (a) peaks for the C=O and C-N stretching modes of ADIC-CSAC at 1767.50–1774.70 and 1324.60–1330.40 cm^{-1} , respectively; however, peaks for C=O and C-N do not appear in the CT-CSAC spectrum; peaks for PhOH, Ph, $-\text{CH}_3$, $-\text{CH}_2$, $-\text{COO}-$, C=S, Ti-O, and

Zn-O correspond to the functional groups contained in each component of C-CSAC; (b) the presence of N fully confirms that Compound C has been incorporated into the coating and the contents of C, O, N, Zn, S, Cu, and Ti are basically the same as that of each component in C-CSAC, among which, the highest contents correspond to C (65.50%) and O (26.70%); (c) EDX element mapping confirms that C-CSAC contains C, O, N, Zn, S, Cu, and Ti and these elements are evenly distributed. The abovementioned results show that ADIC and ADIC-CSAC were successfully synthesized and ADIC was uniformly distributed in ADIC-CSAC.

TG Assay Results. The TG-DTG results for the coatings are shown in Figure 4. The TG-DTG curves of CT-CSAC and ADIC-CSAC exhibit a similar endothermic peak before 800 °C. There is only a small mass loss over Zone-I (which ranges from 30 °C to Temperature 1 (T1)) for the coatings, which could be due to the volatilization of moisture retained by the compounds and the solvent retained by the coatings. There is a mass loss of approximately 50% over Zone-II (which ranges between T1 and Temperature 2 (T2)), where both T1 and T2 of ADIC-CSAC are higher than those of CT-CSAC. The drastic change in the mass over this zone may be caused by the

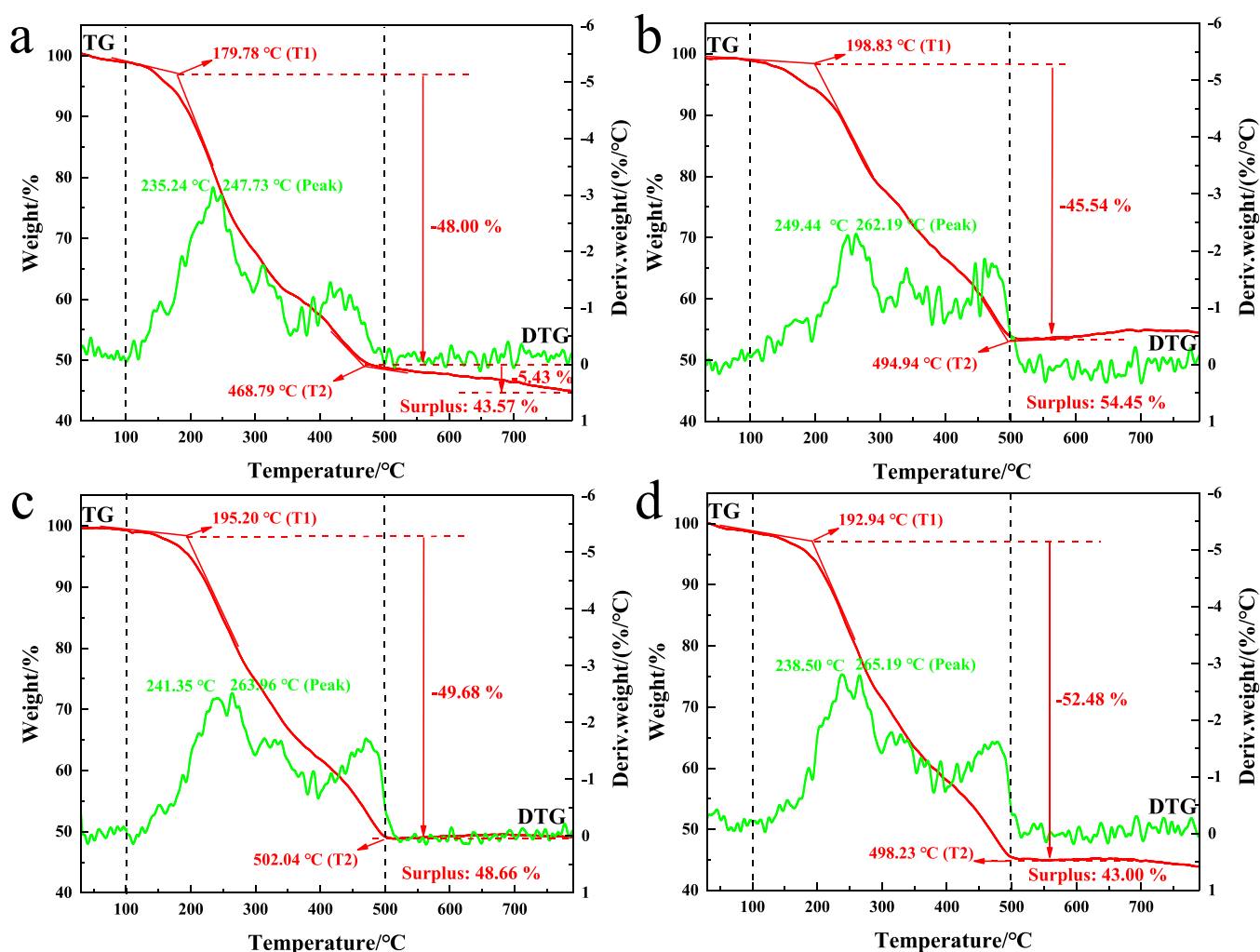


Figure 4. TG-DTG curves of CT-CSAC (a), A-CSAC (b), B-CSAC (c), and C-CSAC (d).

decomposition of the compounds and resins and the failure of the auxiliaries. With an increasing heating rate, the DTG curves of the coatings change significantly. At the end of Zone III, there is a long tail in the DTG curve that terminates at approximately 800 °C, showing that no reactions with the appreciable mass loss take place after T2. The abovementioned conclusions show that all the coatings have good and similar thermal stabilities, where the thermal stability of ADIC-CSAC is slightly superior to that of CT-CSAC. The addition of CT and ADIC has a minimal effect on the stability of the coatings.

Static State Simulation Assay Results. Many previous studies have shown that the antifouling effect of self-polishing antifouling coatings is closely related to the self-polishing performance, roughness, surface wettability, and SE of the coating. Therefore, changes in the mass loss, Ra, WCA, and SE with time were measured in this study, and the results are shown in Figure 5.

The self-polishing performance of the zinc acrylic resin was evaluated in terms of the hydrolysis degradation of the coatings after the static simulation experiment was performed. Figure 5a shows the change in the mass loss of ADIC-CSAC with time after hydrolysis degradation in 25 °C seawater, using the relevant data of CT-CSAC as a control group. First, the results show that the highest mass loss occurred at the initial stage for all the coatings, which may have resulted from the seawater

scouring some solid particles with weak adhesion off the coating surface.⁴⁶ The variation in the mass loss gradually increased during the subsequent experimental periods, which may have resulted from hydrolysis degradation of the zinc acrylic resin, as shown in Figure 5b. The hydrolysis process is realized by ion exchange between the zinc ions of the acrylic resin and sodium ions and chloride ions in seawater. In addition, the structural similarity of A-CSAC, B-CSAC, and C-CSAC resulted in almost the same mass losses for these samples, whereas the mass loss of CT-CSAC was slightly higher than that of ADIC-CSAC, which indicates that CT-CSAC has a better anti-adhesion effect. The incorporation of ADIC and CT slightly affected the hydrolysis degradation process of the zinc acrylate resin but had little impact on the original hydrolysis trend. The hydrolysis of the coatings tended to terminate at 50 days, and the mass loss of all the coatings was less than 8%, indicating the stability of the coatings in seawater. In addition, the Ra, WCA, and SE of the coatings in seawater are summarized in Figure 5c–e. The surface roughness of all the coatings decreased with time, the coatings reached a stable state after the first cycle, and the Ra values were less than 1.52 μm after the fifth cycle. The WCA values of all the coatings were greater than 90 °C, indicating that the coatings were hydrophobic, and the WCA and SE did not change significantly with time over the entire experimental

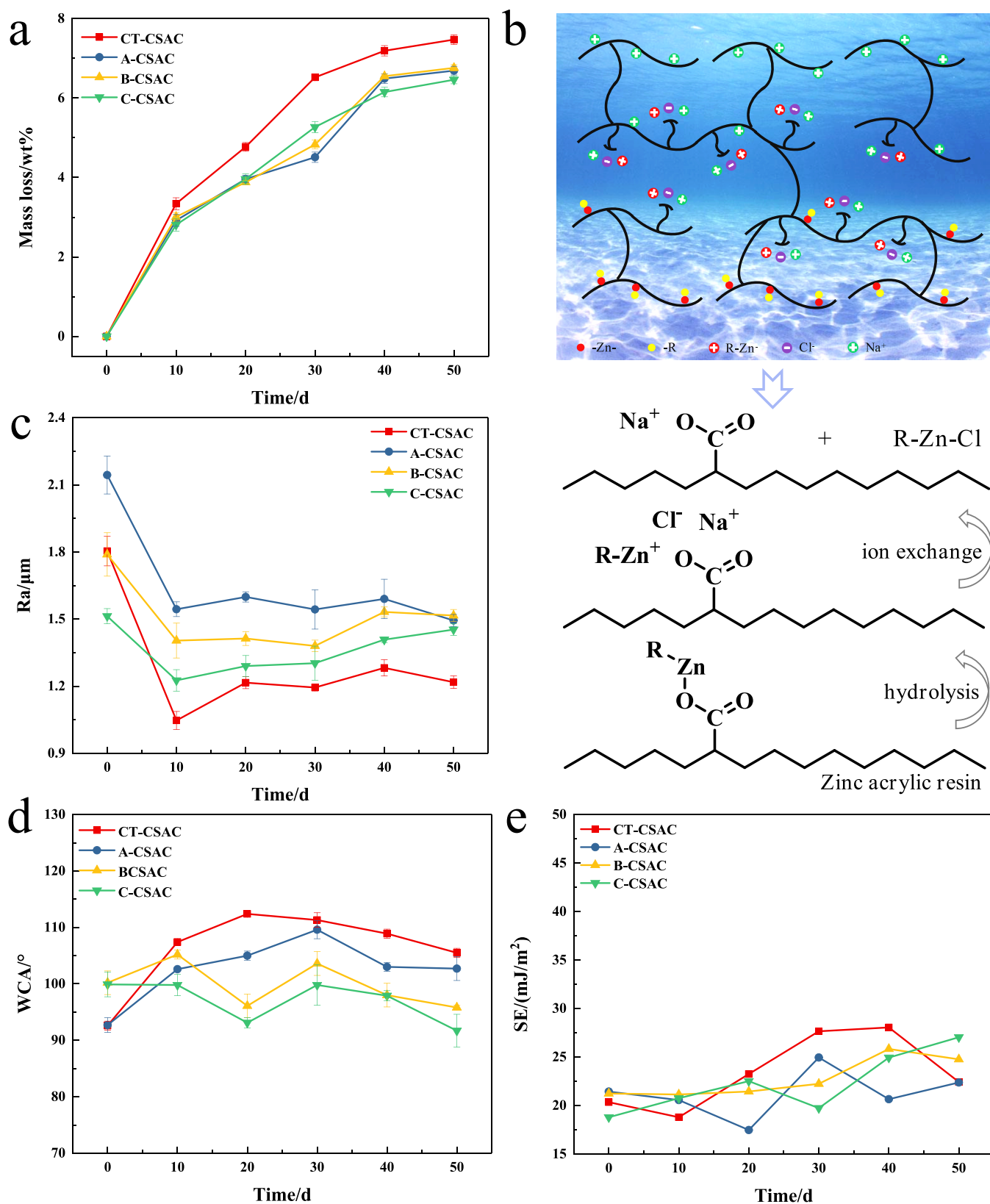


Figure 5. Changes in the mass loss (a), Ra (c), WCA (d), and SE (e) of the coatings and hydrolysis degradation diagram (b) of the zinc acrylic resin in seawater at 25 °C.

process, showing good surface stability for the coatings. The addition of different fungicides slightly changed the surface smoothness and wettability of the coatings. Compared to the properties of ADIC-CSAC, the Ra and SE values of CT-CSAC

were slightly lower and the WCA value was slightly higher. Generally, a good antifouling effect is obtained for a coating with low roughness, a large CA, and a small SE.^{47–49} A preliminary speculation is that CT-CSAC will produce a

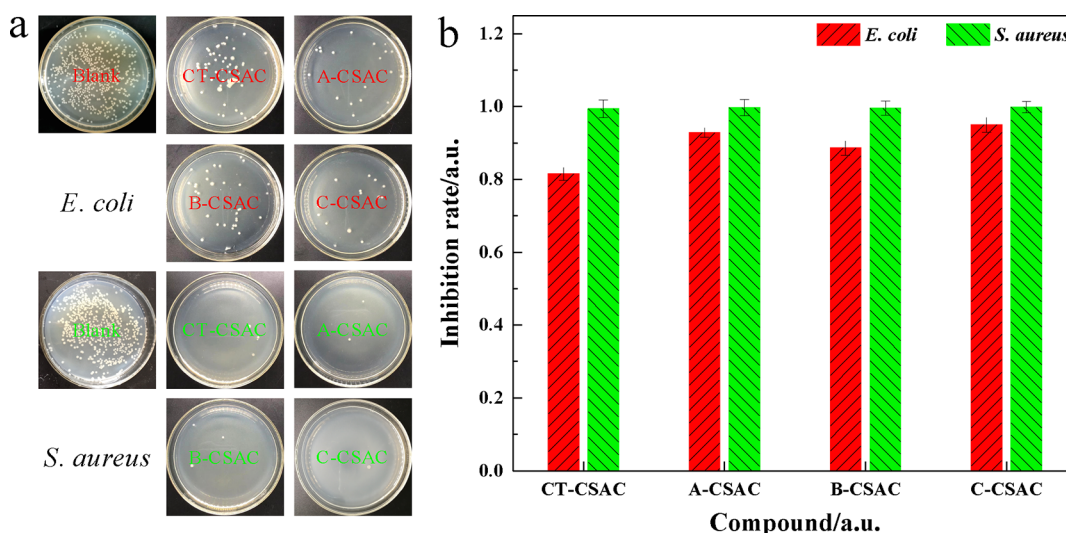


Figure 6. Antibacterial effect (a) and antibacterial rates (b) of the coatings against *E. coli* and *S. aureus*.

slightly better antifouling effect than ADIC-CSAC; however, the final antifouling results should be considered along with the antifouling ability of the coating itself. In addition, the differences in the WCAs of CT-CSAC, A-CSAC, B-CSAC, and C-CSAC may be associated with the different functional groups in CT and ADIC. The benzene ring and chlorine in CT are hydrophobic, whereas the phenolic hydroxyl and amide in ADIC are hydrophilic and can interact with water molecules, thereby driving water motion.⁵⁰ Compounds A, B, and C contain alkene, benzene ring, and methyl groups, respectively, and the difference in the hydrophobicities of these compounds resulted in small differences between the WCAs of A-CSAC, B-CSAC, and C-CSAC. The abovementioned experimental results show that ADIC-CSAC exists in a stable hydrolysis state during the entire hydrolysis process and is therefore likely to have a good antifouling effect.

Antibacterial Test Results. The antibacterial activity of ADIC-CSAC was evaluated against two major classes of bacteria, Gram-negative (*E. coli*) and Gram-positive (*S. aureus*) bacteria. Figure 6 shows that all the coatings had excellent antibacterial effects against both *E. coli* and *S. aureus*, where only a few colonies adhered to CT-CSAC (81.56 and 99.41% inhibition, respectively), A-CSAC (92.91 and 99.71% inhibition, respectively), B-CSAC (88.65 and 99.56% inhibition, respectively), and C-CSAC (95.04 and 99.85% inhibition, respectively). The abovementioned results show that (1) the antibacterial activity of ADIC-CSAC against the tested strains was slightly higher than that of CT-CSAC, where the excellent antibacterial activity of ADIC-CSAC resulted from the synergistic effect of ADIC and the zinc acrylate resin and effectively prevented bacterial adhesion to the coating surface. Upon immersing the coatings in seawater, the zinc acrylate resin begins to hydrolyze, thereby exposing ADIC and Zn^{2+} to seawater, both of which have antibacterial activity. Therefore, the functional groups (amide and phenolic hydroxyl groups) in ADIC change the permeability and induce rupture of the cell membrane, thereby continuously exposing proteins and other substances in the bacterial cell and inhibiting the growth and proliferation of bacterial cells. In addition, the continuous hydrolysis of the zinc acrylate resin enables continual renewal of the coating surface, along with the attached biofilm. (2) CSAC had a higher inhibitory effect on *S. aureus* than on *E.*

coli, which has an additional outermost membrane containing lipopolysaccharides and therefore, additional protection. The results of the antibacterial experiment presented above provide preliminary confirmation of the effectiveness of the coating formulation developed in this study for antifouling.

Anti-Algal Test Results. Four types of diatoms (*P. tricornutum*, *C. curvisetus*, *S. costatum*, and *N. closterium*) were selected to evaluate the anti-algal performance of ADIC-CSAC. The maximum absorption wavelengths of *P. tricornutum*, *C. curvisetus*, *S. costatum*, and *N. closterium* were 683, 682, 683, and 680 nm, respectively, and a good linear relationship was obtained between the algal absorbance and concentration (Table S2). Figure 7 shows that both the experimental and control groups produced excellent algal inhibition effects compared with the blank group, which further confirmed the efficiency of the coating. First, ADIC-CSAC produced an excellent anti-algal effect at an inhibition rate of more than 72%; this effect is compared against that of CT-CSAC below. Compared to the algal inhibition effect of CT-CSAC, that of ADIC-CSAC against *P. tricornutum* was slightly higher, only that of B-CSAC and C-CSAC was higher against *S. costatum*, only that of B-CSAC was higher against *N. closterium*, and that of ADIC-CSAC was slightly lower against *C. curvisetus*. Therefore, among the different types of algae, ADIC-CSAC is the most sensitive to *P. tricornutum* and the least sensitive to *C. curvisetus*. In addition, the algal inhibition effects on *P. tricornutum*, *C. curvisetus*, and *S. costatum* increased in the order of A-CSAC, B-CSAC, and C-CSAC, whereas the algal inhibition effects on *N. closterium* decreased in the order of A-CSAC, B-CSAC, and C-CSAC. In conclusion, ADIC-CSAC can effectively inhibit algal growth and reduce algal adhesion to surfaces, but the degree of inhibition depends on the type of coating and algae.

Marine Field Test Results. Compared to the aforementioned tests, a marine field test is a more direct and effective method for evaluating the antifouling performance of coatings. The sea area of Qingdao Port is characterized by clear regional and seasonal changes and is therefore suitable for investigating the performance of marine antifouling coatings. Figure 8 shows the antifouling effect of ADIC-CSAC after immersion in the sea at Qingdao Port for 9 months. Only a small quantity of the biofilm attached to the boards during the third month. The

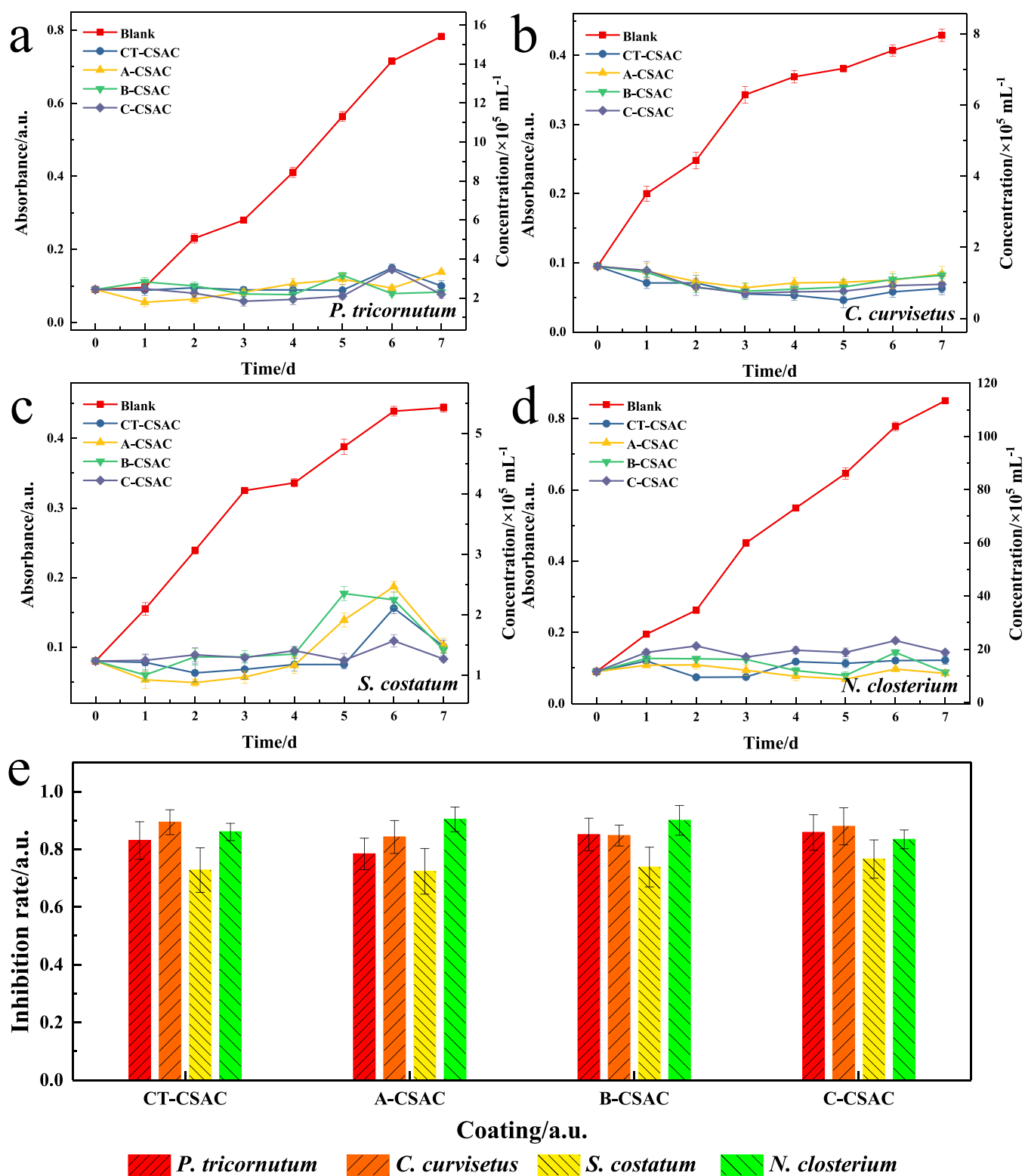


Figure 7. Anti-algal effects of the prepared coatings against *P. tricornutum* (a), *C. curvisetus* (b), *S. costatum* (c), and *N. closterium* (d) and corresponding anti-algal rates on the 7th day of the experiment (e).

growth of marine organisms was slow during the cold season. After the climate warmed up, the marine organisms grew rapidly, such that a large number of marine organisms, such as marine microorganisms, mollusks, and macroalgae, had attached to the blank board by the sixth month. The species of marine organisms became increasingly abundant with time.

Some soil, marine microorganisms, and algae also attached to the control board, and even mollusks grew on the experimental board by the ninth month. Compared to the antifouling effect of CT-CSAC, that of ADIC-CSAC was comparable at the six month and slightly inferior by the ninth month; ADIC-CSAC nevertheless produced a good antifouling effect that was

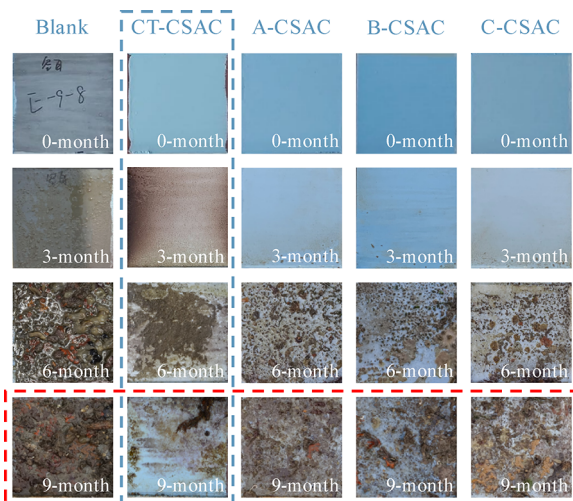


Figure 8. Antifouling effect of ADIC-CSAC over 9 months in Qingdao Port.

consistent with the test results for the self-polishing performance. Early in the experiment, there was no noticeable difference among the antifouling effects of the different ADIC-CSAC coatings, possibly because of the structural similarity of Compounds A, B, and C. However, the antifouling effect of B-CSAC was slightly better than those of A-CSAC and C-CSAC over time, which may be related to the WCA and SE of the coatings. Compared with A-CSAC and C-CSAC, B-CSAC has a smaller WCA and a larger SE, which inhibits the adhesion of marine organisms and improves the antifouling effect. The experimental results presented above demonstrate that the combined action of ADIC and the zinc acrylate resin resulted in good antifouling activity for the coatings, as shown in Figure 9. ADIC effectively inhibits biological growth and

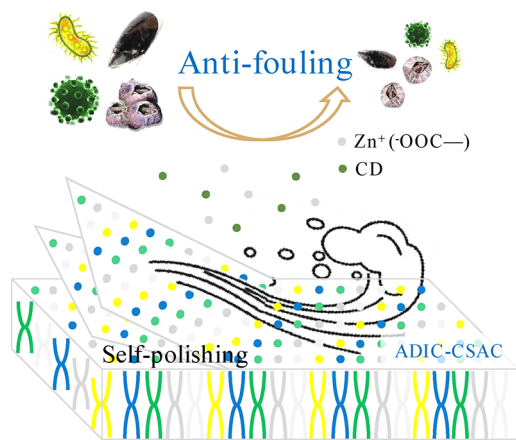


Figure 9. Antifouling mechanism of the prepared coatings in natural seawater.

adhesion. In addition, the main chain of the acrylic resin degrades into small hydrophilic molecules that disperse in seawater, creating a dynamic surface for the coating. The coating design provides further resistance to the initial adhesion of microorganisms and macroorganisms while renewing the antifouling ability or efficacy by separating the outermost severely polluted layer. No surface fracture or falling off of fragments was observed for the coatings during the entire experiment, indicating that the coatings have good mechanical

properties. The integrity of an antifouling coating is critical to obtain an antifouling effect. Good polishing, antifouling, and an ideal morphology are all realized in the prepared coating.

CONCLUSIONS

In summary, several green antifouling ADIC agents were synthesized using salicylamide and amides and evenly added to a coating to prepare a series of CSAC that are collectively labeled as ADIC-CSAC. The as-prepared ADIC-CSAC effectively prevented the adhesion of *E. coli*, *S. aureus*, *P. tricornutum*, *C. curvictus*, *S. costatum*, and *N. closterium*, exhibiting superior antifouling to CSAC in a natural marine environment. This result was obtained because of the synergy between the excellent self-polishing properties of the zinc acrylic resin and the ability of ADIC to inhibit biological attachment, and the slow and steady hydrolysis of ADIC-CSAC produced a good antifouling effect. Moreover, the as-prepared ADIC-CSAC possesses good wear resistance and durability under marine conditions. In short, the prepared ADIC-CSAC has outstanding antifouling and self-polishing properties and is environmentally friendly because of being completely Cu₂O-free. The development of this coating lays a foundation for the preparation of environmentally friendly, lasting, and good antifouling effect coatings, while significantly reducing copper emissions and playing a very important role in protecting the marine ecological environment.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.2c01503>.

IR, ¹H NMR, ¹³C NMR, and HRMS of A; IR, ¹H NMR, ¹³C NMR, and HRMS of B; surface energy parameters of water and diiodomethane; the algal maximum absorption wavelengths (MAW), standard working equations (SWE), and correlation coefficient (CC) of the algal absorbance and concentration (PDF)

AUTHOR INFORMATION

Corresponding Authors

Xiaohui Jiang — Sanya Oceanographic Institution, Ocean University of China, Sanya 572024, China; Frontiers Science Center for Deep Ocean Multispheres and Earth System, Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China; Open Studio for Marine Corrosion and Protection, Pilot National Laboratory for Marine Science and Technology (Qingdao), Qingdao 266100, China; orcid.org/0000-0002-8343-3447; Email: jiangxiaohui123457@163.com

Liangmin Yu — Sanya Oceanographic Institution, Ocean University of China, Sanya 572024, China; Frontiers Science Center for Deep Ocean Multispheres and Earth System, Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China; Open Studio for Marine Corrosion and Protection, Pilot National Laboratory for Marine Science and Technology (Qingdao), Qingdao 266100, China; Email: yuyan@ouc.edu.cn

Authors

Xuan Wang – Sanya Oceanographic Institution, Ocean University of China, Sanya 572024, China; Frontiers Science Center for Deep Ocean Multispheres and Earth System, Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China

Jian Yang – Frontiers Science Center for Deep Ocean Multispheres and Earth System, Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China

Zhenxia Liu – Sanya Oceanographic Institution, Ocean University of China, Sanya 572024, China; Frontiers Science Center for Deep Ocean Multispheres and Earth System, Key Laboratory of Marine Chemistry Theory and Technology, Ministry of Education, Ocean University of China, Qingdao 266100, China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.langmuir.2c01503>

Author Contributions

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Verhougstraete, M. P.; Martin, S. L.; Kendall, A. D.; Hyndman, D. W.; Rose, J. B. Linking fecal bacteria in rivers to landscape, geochemical, and hydrologic factors and sources at the basin scale. *Proc. Natl. Acad. Sci.* **2015**, *112*, 10419–10424.
- (2) Qin, L.; Hafezi, M.; Yang, H.; Dong, G.; Zhang, Y. Constructing a dual-function surface by microcasting and nanospraying for efficient drag reduction and potential antifouling capabilities. *Micromachines* **2019**, *10*, 490.
- (3) Yang, W.; Lin, P.; Cheng, D.; Zhang, L.; Wu, Y.; Liu, Y.; Pei, X.; Zhou, F. Contribution of charges in polyvinyl alcohol networks to marine antifouling. *ACS Appl. Mater. Interfaces* **2017**, *9*, 18295–18304.
- (4) Hader, D. P.; Banaszak, A. T.; Villafane, V. E.; Narvarte, M. A.; Gonzalez, R. A.; Helbling, E. W. Anthropogenic pollution of aquatic ecosystems: Emerging problems with global implications. *Sci. Total Environ.* **2020**, *713*, No. 136586.
- (5) Sharma, S.; Chatterjee, S. Microplastic pollution, a threat to marine ecosystem and human health: a short review. *Environ. Sci. Pollut. R.* **2017**, *24*, 21530–21547.
- (6) Avio, C. G.; Gorbi, S.; Regoli, F. Plastics and microplastics in the oceans: from emerging pollutants to emerged threat. *Mar. Environ. Res.* **2017**, *128*, 2–11.
- (7) Wang, P.; He, B. L.; Wang, B. W.; Wang, L.; Yu, H.; Liu, S. J.; Ye, Q.; Zhou, F. Durable self-polishing antifouling coating based on fluorine-containing pyrrolidone amphiphilic copolymer-functionalized nanosilica. *Prog. Org. Coat.* **2022**, *165*, No. 106706.
- (8) Zhang, J. B.; Wang, X. W.; Zhang, C. Y.; Yan, F. H.; Yu, B.; Yang, W. F.; Pei, X. W.; Zhou, F. Self-lubricating interpenetrating polymer networks with functionalized nanoparticles enhancement for quasi-static and dynamic antifouling. *Chem. Eng. J.* **2022**, *429*, No. 132300.
- (9) Yebra, D. M.; Kiil, S.; Dam-Johansen, K. Antifouling technology—past, present and future steps towards efficient and environmentally friendly antifouling coatings. *Prog. Org. Coat.* **2004**, *50*, 75–104.
- (10) Long, Y.; Yu, Y. H.; Yin, X.; Li, J.; Carlos, C.; Du, X. S.; Jiang, Y. D.; Wang, X. D. Effective anti-biofouling enabled by surface electric disturbance from water wave-driven nanogenerator. *Nano Energy* **2019**, *57*, 558–565.
- (11) Perez, M.; Sanchez, M.; Garcia, M.; Patino, C. L.; Blustein, G.; Palermo, J. A. Antifouling activity of peracetylated cholic acid, a natural bile acid derivative. *Steroids* **2019**, *149*, 108414.
- (12) Mirmohseni, A.; Azizi, M.; Seyed Dorraji, M. S. Facile synthesis of copper/reduced single layer graphene oxide as a multifunctional nanohybrid for simultaneous enhancement of antibacterial and antistatic properties of waterborne polyurethane coating. *Prog. Org. Coat.* **2019**, *131*, 322–332.
- (13) Sae-ung, P.; Kolewe, K. W.; Bai, Y.; Rice, E. W.; Schiffman, J. D.; Emrick, T.; Hoven, V. P. Antifouling stripes prepared from clickable zwitterionic copolymers. *Langmuir* **2017**, *33*, 7028–7035.
- (14) Xie, Q. Y.; Pan, J. S.; Ma, C. F.; Zhang, G. Z. Dynamic surface antifouling: Mechanism and systems. *Soft Matter* **2019**, *15*, 1087–1107.
- (15) Xu, G.; Liu, P.; Pranantyo, D.; Neoh, K. G.; Kang, E. T. Dextran- and chitosan-based antifouling, antimicrobial adhesion, and self-polishing multilayer coatings from pH-responsive linkages-enabled layer-by-layer assembly. *ACS Sustainable Chem. Eng.* **2018**, *6*, 3916–3926.
- (16) Kwon, S. H.; Lee, I.; Park, H.; Lee, S. G. Decomposition mechanisms of self-polishing copolymers for antifouling coating materials through first-principles approach. *Prog. Org. Coat.* **2020**, *138*, No. 105406.
- (17) Yang, H. J.; Chang, H.; Zhang, Q.; Song, Y. Y.; Jiang, L.; Jiang, Q. M.; Xue, X. Q.; Huang, W. Y.; Ma, C. F.; Jiang, B. B. Highly branched copolymers with degradable bridges for antifouling coatings. *ACS Appl. Mater. Interfaces* **2020**, *12*, 16849–16855.
- (18) Kiaune, L.; Nan, S. Pesticidal copper (I) oxide: Environmental fate and aquatic toxicity. *Rev. Environ. Contam. Toxicol.* **2011**, *213*, 1–26.
- (19) Liu, Y.; Suo, X. K.; Wang, Z.; Gong, Y. F.; Wang, X.; Li, H. Developing polyimide-copper antifouling coatings with capsule structures for sustainable release of copper. *Mater. Design* **2017**, *130*, 285–293.
- (20) Wu, W. T.; Zhao, W. J.; Wu, Y. H.; Zhou, C. X.; Li, L. Y.; Liu, Z. X.; Dong, J. D.; Zhou, K. H. Antibacterial behaviors of Cu₂O particles with controllable morphologies in acrylic coatings. *Appl. Surf. Sci.* **2019**, *465*, 279–287.
- (21) Dai, G. X.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Biodegradable poly(ester-co-acrylate) with antifoulant pendant groups for marine anti-biofouling. *ACS Appl. Mater. Interfaces* **2019**, *11*, 11947–11953.
- (22) Ciriminna, R.; Bright, F. V.; Pagliaro, M. Ecofriendly antifouling marine coatings. *ACS Sustainable Chem. Eng.* **2015**, *3*, 559–565.
- (23) Sun, B. K.; Fan, H. S.; Pan, X. L.; Lu, A. D.; Hu, J. K. Development of copper-free self-polishing anti-fouling paints by using acrylate resin. *Surf. Tech.* **2022**, *51*, 280–286.
- (24) Heine, L.; Nestler, A. Promising practices for alternatives assessment: Lessons from a case study of copper-free antifouling coatings. *Integr. Environ. Assess. Manage.* **2019**, *15*, 867–879.
- (25) Edwards, C. D.; Pawluk, K. A.; Cross, S. F. The effectiveness of several commercial antifouling treatments at reducing biofouling on finfish aquaculture cages in British Columbia. *Aquac. Res.* **2015**, *46*, 2225–2235.
- (26) Jerabek, A. S.; Wall, K. R.; Stallings, C. D. A practical application of reduced-copper antifouling paint in marine biological research. *PeerJ* **2016**, *4*, No. e2213.
- (27) Faÿ, F.; Gouessan, M.; Linossier, I.; Réhel, K. Additives for efficient biodegradable antifouling paints. *Int. J. Mol. Sci.* **2019**, *20*, 361–372.

- (28) Sen, K.; Erdogan, U. H.; Cavas, L. Prevention of biofouling on aquaculture nets with eco-friendly antifouling paint formulation. *Color. Technol.* **2020**, *136*, 120–129.
- (29) Zhao, X. K.; Bai, X. Q.; Yuan, C. Q. Progress in green biological antifoulant and controlled release technologies. *Ship Sci. Tech.* **2017**, *39*, 6–11.
- (30) Liu, Y. L.; Wang, S. L.; Li, C. G.; Fang, D. Q.; Wu, J. H. Preparation and performance of environment-friendly copper-free self-polishing antifouling paints. *Surf. Tech.* **2017**, *46*, 1–5.
- (31) Wang, X.; Jiang, X. H.; Yu, L. M. Performance assessment of capsaicin derivatives containing amide groups used as active substances for antifouling coatings. *Prog. Org. Coat.* **2021**, *160*, No. 106515.
- (32) Yang, X. L.; Li, Y.; Lei, W. W.; Liu, X. D.; Zeng, Q.; Liu, Q. C.; Feng, W.; Li, K.; Wang, P. Thermal degradation behaviors of poly(arylene ether nitrile) bearing pendant carboxyl groups. *Polym. Degrad. Stab.* **2021**, *191*, No. 109668.
- (33) Antunes, L. R.; Breitenbach, G. L.; Pellá, M. C. G.; Caetano, J.; Dragunski, D. C. Electrospun poly(lactic acid) (PLA)/poly(butylene adipate-co-terephthalate) (PBAT) nanofibers for the controlled release of cilostazol. *Int. J. Biol. Macromol.* **2021**, *182*, 333–342.
- (34) Yilmaz, H.; Cobandede, Z.; Yilmaz, D.; Cinkilic, A.; Culha, M.; Demiralay, E. C. Monitoring forced degradation of drugs using silica coated AgNPs with surface-enhanced Raman scattering. *Talanta* **2020**, *214*, No. 120828.
- (35) Gevaux, L.; Lejars, M.; Margailan, A.; Briand, J. F.; Bunet, R.; Bressy, C. Hydrolyzable additive-based silicone elastomers: A new approach for antifouling coatings. *Polymer* **2019**, *11*, 305.
- (36) Wei, D. W.; Wei, H. Y.; Gauthier, A. C.; Song, J. L.; Jin, Y. C.; Xiao, H. N. Superhydrophobic modification of cellulose and cotton textiles: Methodologies and applications. *J. Bioresour. Bioprod.* **2020**, *5*, 1–15.
- (37) Han, X. T.; Gong, X. In situ, one-pot method to prepare robust superamphiphobic cotton fabrics for high buoyancy and good antifouling. *ACS Appl. Mater. Interfaces* **2021**, *13*, 31298–31309.
- (38) Guan, F. X.; Song, Z. P.; Xin, F. R.; Wang, H. L.; Yu, D. H.; Li, G. D.; Liu, W. X. Preparation of hydrophobic transparent paper via using polydimethylsiloxane as transparent agent. *J. Bioresour. Bioprod.* **2020**, *5*, 37–43.
- (39) Zheng, X.; Yu, H. Y.; Gong, X. Designing photothermal superhydrophobic PET fabrics via in situ polymerization and 1,4-conjugation addition reaction. *Langmuir* **2022**, *38*, 8707–8718.
- (40) Scilletta, N. A.; Pezzoni, M.; Desimone, M. F.; Soler-Illia, G. J. A. A.; Bellino, M. G.; Catalano, P. N. Determination of antibacterial activity of film coatings against four clinically relevant bacterial strains. *Bio-Protoc.* **2021**, *11*, No. e3887.
- (41) Song, L.; Xiao, Y. F.; Gan, L.; Wu, Y.; Wu, F.; Gu, Z. W. The effect of antibacterial ingredients and coating microstructure on the antibacterial properties of plasma sprayed hydroxyapatite coatings. *Surf. Coat. Technol.* **2012**, *206*, 2986–2990.
- (42) Du, J.; Li, Z. Y.; Li, Z. Y.; Guo, H. Y.; Zhu, E. W.; Liu, C. B.; Ren, B. X.; Che, G. B. The facile preparation and antibacterial performance of a conductive polymer-PU coating under visible light. *Prog. Org. Coat.* **2022**, *165*, No. 106755.
- (43) Zhou, W. J.; Wang, Y. J.; Ni, C. H.; Yu, L. M. Preparation and evaluation of natural rosin-based zinc resins for marine antifouling. *Prog. Org. Coat.* **2021**, *157*, No. 106270.
- (44) Zhou, W. J.; Zhou, Y.; Ni, C. H.; Yu, L. M.; Li, C. C.; Yan, X. F.; Zhang, X. H. Research on electromagnetic wave absorption properties of zinc-based acrylate resins for marine antifouling coating. *J. Alloys Compd.* **2022**, *900*, No. 163285.
- (45) Wang, X.; Jiang, X. H.; Yu, L. M. Preparation and evaluation of polyphenol derivatives as potent antifouling agents: addition of a side chain affects the biological activity of polyphenols. *Biofouling* **2022**, *38*, 29–41.
- (46) Ni, C. H.; Feng, K.; Li, X.; Zhao, H. Z.; Yu, L. M. Study on the preparation and properties of new environmentally friendly antifouling acrylic metal salt resins containing indole derivative group. *Prog. Org. Coat.* **2020**, *148*, No. 105824.
- (47) Schmidt, D. L.; Robert, F., Jr.; Brady, L.; Karen, D. C.; Schmidt Chaudhury, M. K. Contact angle hysteresis, adhesion, and marine biofouling. *Langmuir* **2004**, *20*, 2830–2836.
- (48) Lakshmi, K.; Muthukumar, T.; Doble, M.; Vedaprakash, L.; Kruparathnam; Dineshram, R.; Jayaraj, K.; Venkatesan, R. Influence of surface characteristics on biofouling formed on polymers exposed to coastal sea waters of India. *Colloids Surf., B* **2012**, *91*, 205–211.
- (49) Wang, Y. P.; Pitet, L. M.; Finlay, J. A.; Brewer, L. H.; Cone, G.; Betts, D. E.; Callow, M. E.; Callow, J. A.; Wendt, D. E.; Hillmyer, M. A.; DeSimone, J. M. Investigation of the role of hydrophilic chain length in amphiphilic perfluoropolyether/poly(ethylene glycol) networks: towards high-performance antifouling coatings. *Biofouling* **2011**, *27*, 1139–1150.
- (50) Liu, X. L.; Xu, Y. J.; Wu, Z. Q.; Chen, H. Poly(N-vinylpyrrolidone)-modified surfaces for biomedical applications. *Macromol. Biosci.* **2013**, *13*, 147–154.



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